Backend Workshop

Client–Server, HTTP, APIs, and Databases

2026-04-07 • Beginners to early-intermediate developers

Catalog

1 Workshop outcomes

2 Backend fundamentals

3 Client–server & request lifecycle

4 Internet & HTTP

5 Methods, status codes & payload formats

6 Debugging with DevTools

7 APIs & REST design

8 Architectures, tech stack & data storage

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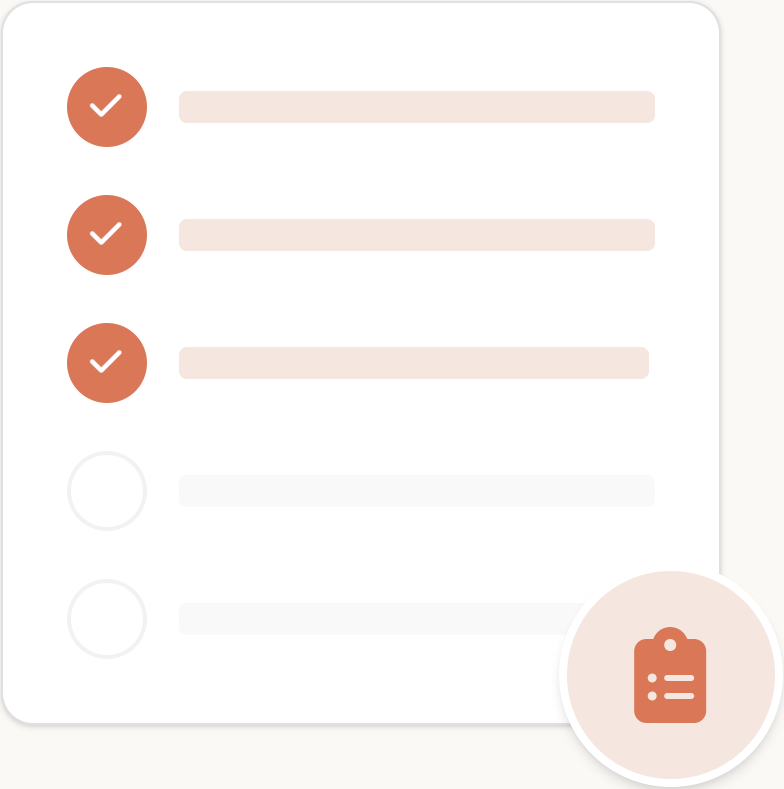
What You Will Learn

By the end of this workshop, you will be able to:

- Understand how web and mobile applications communicate with backend systems
- Explain the request–response lifecycle
- Use browser DevTools to inspect and debug API calls
- Design and interact with REST APIs
- Understand HTTP methods and status codes
- Work with data formats (JSON, XML)
- Distinguish server-rendered apps vs modern SPA architectures
- Build a basic backend service backed by a database
- Explain (at a high level) how backend systems are deployed and scaled



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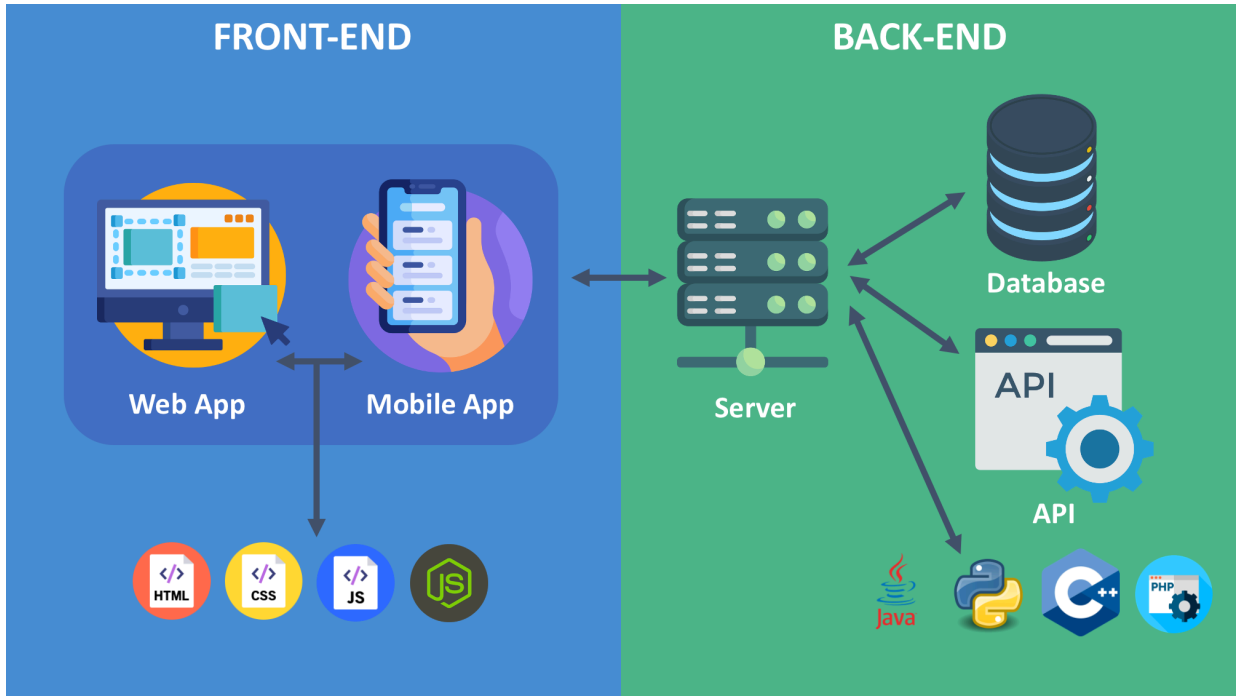
What is Backend?

Definition: The backend is the server-side part of an application that powers functionality and data.

Components: Client → API Layer → Server App → Database

Handles:

- Data storage & retrieval
- Business logic (rules, validations)
- Authentication & authorization
- API communication with clients



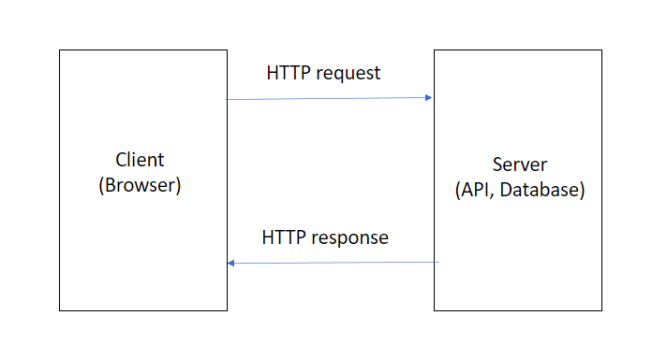
Client vs Server

Client (Browser / Mobile app)

- Sends requests (e.g., fetch profile, submit form)
- Renders UI and reacts to responses

Server

- Receives requests
- Executes logic
- Reads/writes data
- Sends responses



Key concept: Communication happens over the internet using HTTP/HTTPS

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The Lifecycle of a Web Request

Request = input

(what the client asks for)

Response = output

(what the server returns)

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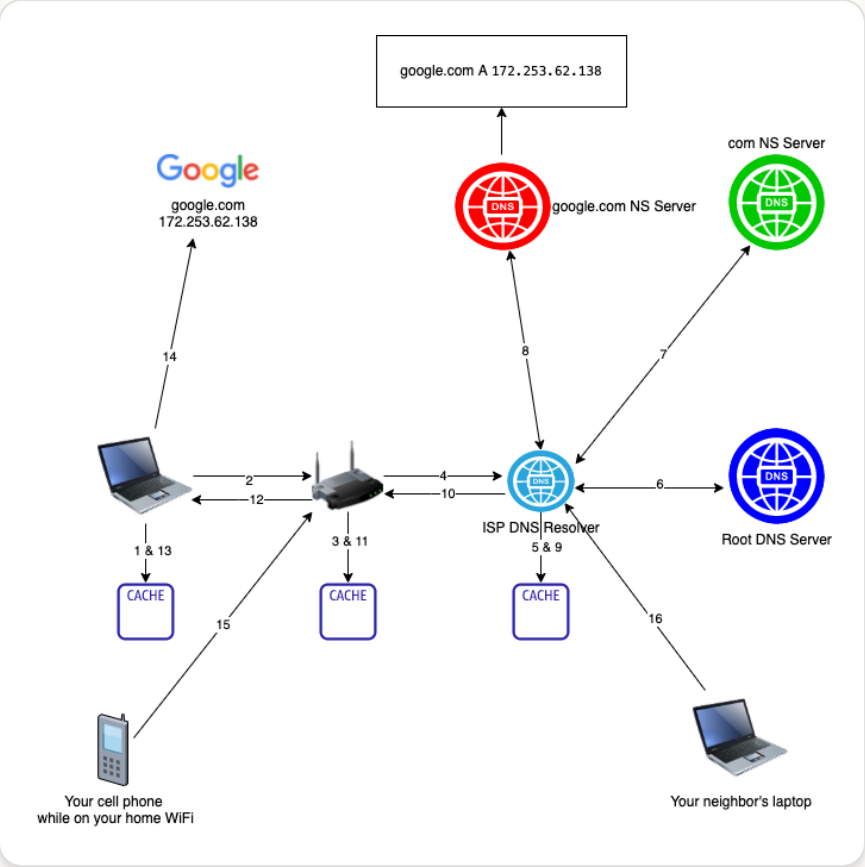
How the Internet Works

Key concepts:

- Domain → IP (DNS lookup)
- Communication over HTTP/HTTPS
- Servers are addressed by IP + port

Ports:

80	→ HTTP	443	→ HTTPS



What an HTTP Message Contains

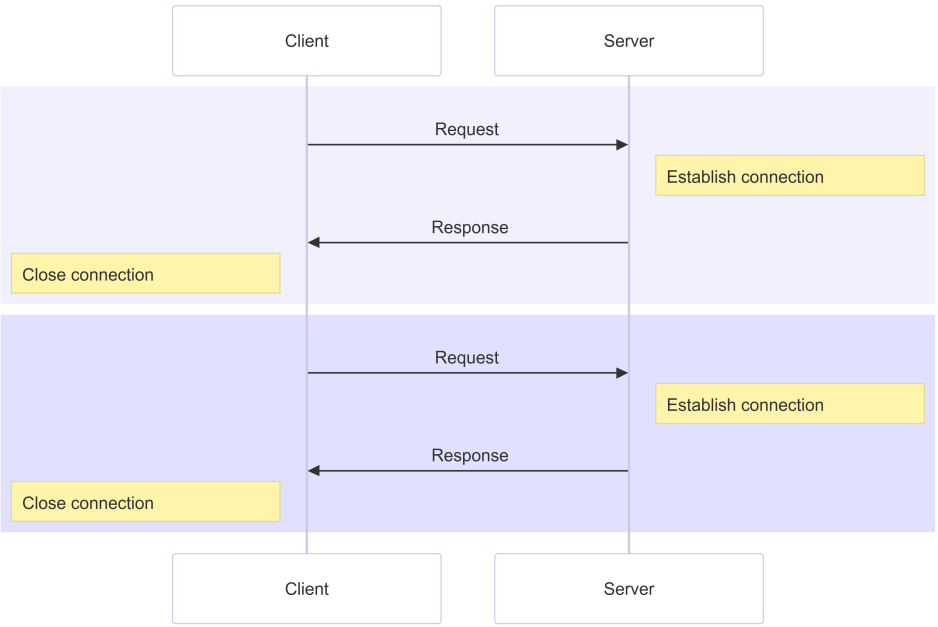
Request Structure

- Method
- URL
- Headers
- Body *(optional)*

Response Structure

- Status code
- Headers
- Body

Stateless: Each request is independent; the server does not remember prior requests by default.



Verbs That Define the Action

GET

Retrieve data

```
e.g., GET /users
```

POST

Create resource

```
e.g., POST /orders
```

PUT

Update/replace resource

```
e.g., PUT /users/123
```

PATCH

Partial update

```
e.g., PATCH /users/123
```

DELETE

Remove resource

```
e.g., DELETE /orders/1
```

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How Servers Communicate Outcomes

2xx
Success

4xx
Client errors

5xx
Server errors

Common Codes

200	OK	401	Unauthorized
201	Created	404	Not Found
400	Bad Request	500	Internal Server Error

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What Data Looks Like Over the Wire

JSON (standard, lightweight)

```
{
  "user": "yousri",
  "email": "test@mail.com"
}
```

XML (legacy, verbose)

```
<user>
  <name>yousri</name>
</user>
```

- Used for API communication
- Serialization: object → text
- Deserialization: text → object

Browser DevTools

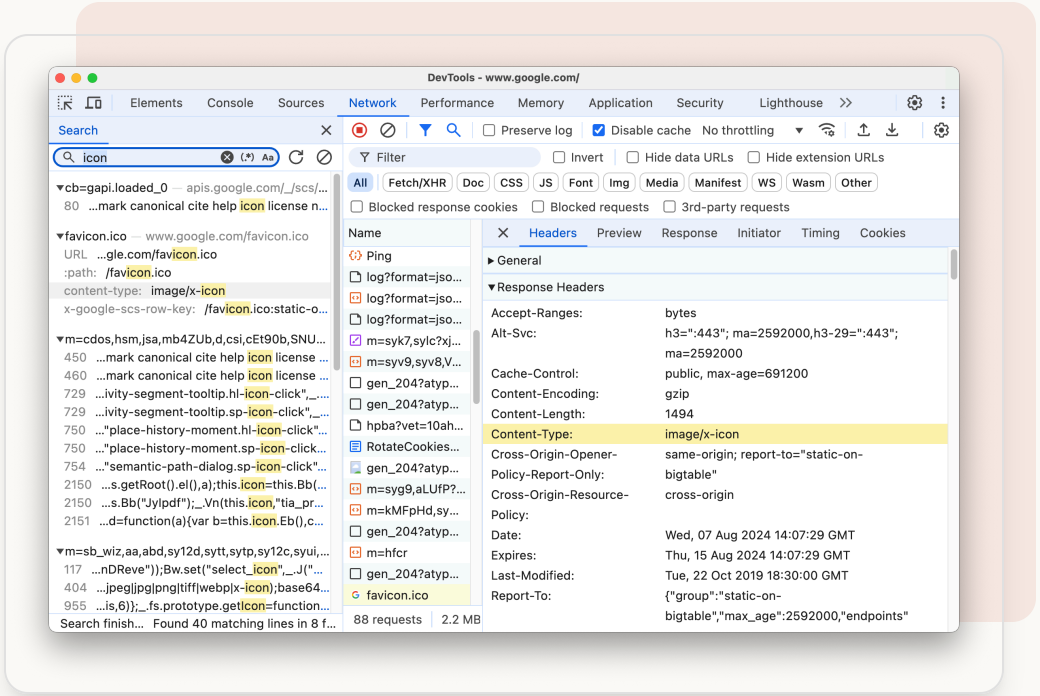
Inspect and Debug API Calls

What to look at

- Request URL (endpoint)
- Request headers
- Request payload (if any)
- Response body (data)
- Status code

How to use it (Demo flow)

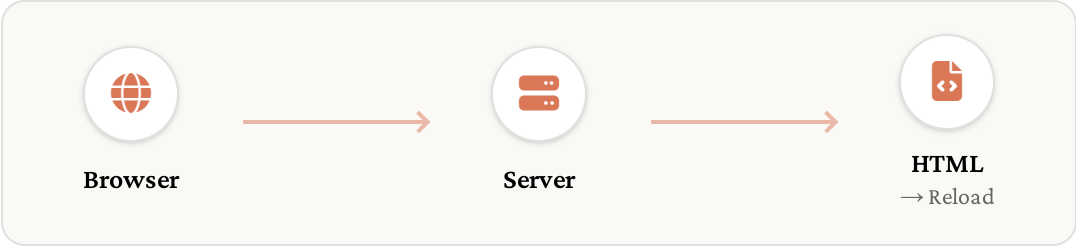
1. Open any site
2. Trigger an action
3. Select an `/api/...` request
4. Inspect details



SPA vs Traditional Applications

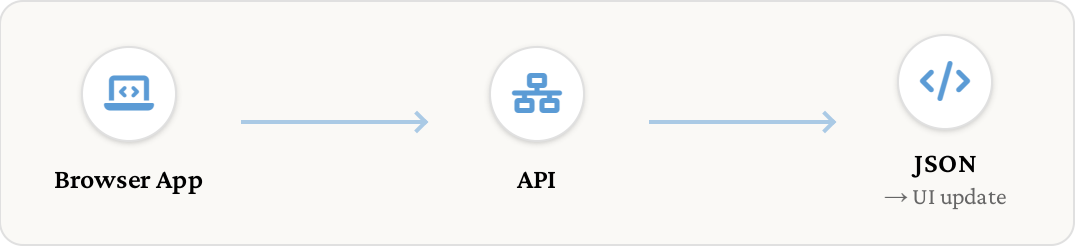
Traditional (server-rendered, e.g., PHP)

- Server returns full HTML pages
- Each request reloads page
- Backend handles rendering



SPA (React / Angular)

- Frontend loads once
- Backend returns JSON
- UI updates dynamically



Key difference: Server → HTML (traditional) vs Server → JSON (SPA)

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Common Server-Side Tools

Languages

- JavaScript (Node.js)
- Python
- Java
- Ruby

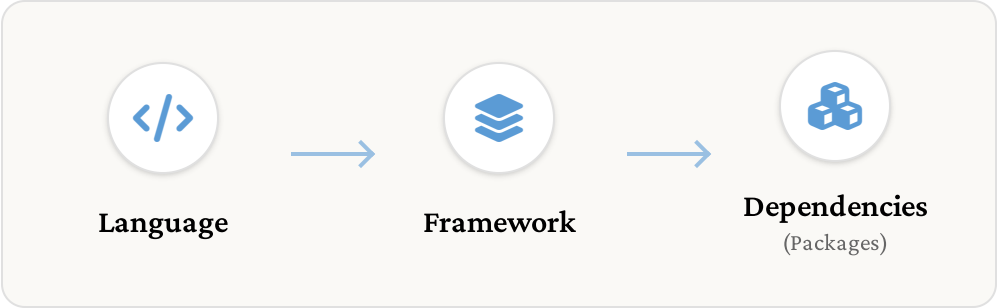
Frameworks

- Express (Node.js)
- Django / FastAPI (Python)
- Spring (Java)

Package Managers

- NPM
- PIP
- Maven

SIMPLE STACK ARCHITECTURE



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Databases

Where Data Lives

Purpose

Store and manage application data securely and efficiently for retrieval and updates.

Types

SQL

Relational, structured tables

PostgreSQL | MySQL

NoSQL

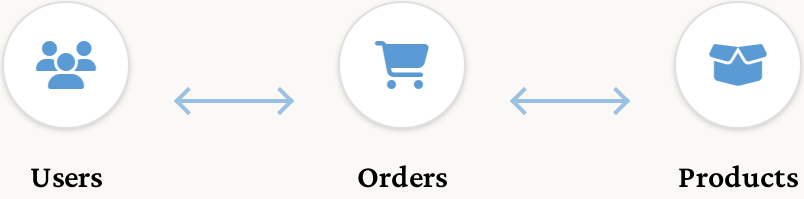
Flexible, document-based

MongoDB

Examples of Stored Data

- Users
- Orders
- Products

MINI DATA MODEL



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Let's Practice

Q&A